WEEK 8 LAB

Hormones and the autonomic system.

2 to 3 hours. Graded. Investigate It, 25 percent of your grade across the term.

This week has five short parts. You watch your own autonomic nervous system change your heart rate. You run three PhysioEx endocrine activities, including one where you diagnose patients from two hormone levels. You run the ELISA and read an unknown off a standard curve. You read one menstrual cycle from its hormone graphs. And one clinical case ties it together.

Write every answer in your own words

1 min

Do not copy from the book, the notes, a website, a classmate or an AI tool, even if you change a few words. Read the source, look away, and then write what you understood. If you cannot say it without looking, that tells you what to go back and study. Your own words, even rough ones, are what I want to see.

What you need

1 min

- PhysioEx, from the Pearson link in Canvas. You use Exercise 4, Endocrine System Physiology, and Exercise 12, Serological Testing.
- For the heart rate test at home: a timer, a place to lie down, and your own pulse at the wrist or the side of the neck. A phone or watch that shows heart rate helps but is not needed.
- The hormone cycle sheet for Part 4, printed or open on screen, and a pencil. Paper and a pen for your standard curve, or the printed worksheet. You can type everything else on this page.

What goes in each box

1 min

1. **What did you measure?** Name what you changed and what you recorded, with units.
2. **What were your findings?** Give the actual results, with numbers. What went up, what went down, and what stayed the same.
3. **What did you learn?** Explain why the results came out that way, using the physiology from this week. This is the most important box.

Your worksheet is graded once PhysioEx shows each activity complete. The PhysioEx Lab Report that Pearson makes is not this worksheet and earns no points.

Part 1. Your autonomic nervous system, at home

1 min

Heart rate with breathing, and with standing up

2 min

Your heart rate is set from moment to moment by the two branches of the autonomic nervous system. These two tests let you watch them work. Find your pulse at the thumb side of your wrist, or at the side of your neck, with two fingers, not your thumb.

Test A. Breathing

1. Sit quietly for 5 minutes.
2. Count your pulse for 30 seconds while breathing normally, and double it. Do this twice and write both.
3. Now breathe slowly, about 5 seconds in and 5 seconds out, for one minute, with your fingers on your pulse. Notice when the beats come closer together and when they spread apart. If a phone or watch shows your heart rate, read the highest number while you breathe in and the lowest while you breathe out, for three breaths.

BREATHING TEST	YOUR RESULT
Resting heart rate, first count (beats per minute)	
Resting heart rate, second count (beats per minute)	
During slow breathing: faster on the breath in or the breath out?	
Highest and lowest rates, if you have a phone or watch reading	

Test B. Standing up

1. Lie flat for 5 minutes. Count your pulse for 30 seconds and double it.
2. Stand up and stay still. Count your pulse for 30 seconds starting at 1 minute, and again starting at 3 minutes.

If you feel dizzy or light-headed, sit or lie back down right away. Skip Test B if you have ever fainted on standing up or have a heart condition, and say so on your worksheet. This is practice of a test, not a medical test.

STANDING TEST	HEART RATE (BEATS PER MINUTE)
Lying down, after 5 minutes	
Standing, at 1 minute	
Standing, at 3 minutes	

In Test A, your heart rate rose and fell with your breathing within a beat or two. Which branch of the autonomic nervous system was being turned down and back up to do that, and how does the speed of the change tell you?

In Test B, what happened to your heart rate when you stood, with numbers? Standing lets blood pool in your legs. Trace the reflex: what sensed the change, which branch took over, and which receptor on the heart carried out the change.

Which receptor on the blood vessels helped you stay standing without fainting, and what did it make the vessels do?

Part 2. PhysioEx Exercise 4, Endocrine System Physiology

1 min

Run these three activities, in the order PhysioEx lists them. Pearson sometimes renumbers activities, so go by the topic named below. The activity on metabolism and thyroid hormone is not part of this week; you do it in Week 12, with metabolism.

Activity 1

1 min

Plasma glucose, insulin, and diabetes mellitus

What did you measure?

What were your findings?

What did you learn?

Draw or describe the glucose tolerance curves you made, with the starting glucose marked on each. Which curve fits diabetes, and what feature of the curve told you?

Activity 2

1 min

Hormone replacement therapy

What did you measure?

What were your findings?

What did you learn?

Which treatment kept bone density closest to normal after the ovaries were removed? Use your numbers.

Activity 3

1 min

Measuring cortisol and adrenocorticotropic hormone (ACTH)

What did you measure?

What were your findings?

What did you learn?

PATIENT	CORTISOL: HIGH, NORMAL OR LOW	ACTH: HIGH, NORMAL OR LOW	YOUR DIAGNOSIS, AND PRIMARY OR SECONDARY
Patient 1			
Patient 2			
Patient 3			
Patient 4			
Patient 5			

Use as many rows as PhysioEx gives you patients.

Pick one patient whose two levels moved in opposite directions and one whose levels moved in the same direction. Explain how negative feedback produces each pattern, and where in the axis each problem must be.

Part 3. Measuring a hormone you cannot see

1 min

Activity 4

1 min

PhysioEx Exercise 12, the ELISA activity

What did you measure?

What were your findings?

What did you learn?

In one or two sentences, what does the color in a well tell you, and what has to be stuck to the well for any color to appear?

Read an unknown off a standard curve

1 min

A hormone assay works like the ELISA you just ran, but it measures how much hormone is in a sample. You first run standards, samples with known amounts of hormone, and record the color each one makes as a number called absorbance. The line through those points is the standard curve. Then you read your patient's sample off it. Below are practice standards for an assay of thyroid-stimulating hormone (TSH), where more hormone gives more color.

TSH IN THE STANDARD (MIU PER L)	ABSORBANCE
0	0.05
0.5	0.15
1	0.25
2	0.45
5	1.05
10	1.85

Draw it on paper: Your standard curve, by hand: TSH on the x axis and absorbance on the y axis. Plot all six standards and draw the line through them. Then draw the guide lines you use to read each unknown.

Sample A gives an absorbance of 0.65. Read its TSH off your curve. Show the guide lines on your drawing and write the value here.

Sample B gives an absorbance of 2.40, above your highest standard. What should the lab do with sample B, and why can you not just extend the line?

The usual reference range for TSH is about 0.4 to 4.0 mIU per L. Is sample A in range? What is the one more test you would want before saying anything about this person's thyroid?

Part 4. Reading the hormone cycle

1 min

The hormone cycle sheet

1 min

Open the hormone cycle sheet. It has two graphs of one 28 day cycle and ten questions. Work it on paper: print it, or copy the graphs onto your own paper. You will mark the day of ovulation, label the phases, and redraw one curve, so it has to be done by hand. Photograph both pages and add them to the PDF you upload.

After you finish the sheet: which hormone peak did you use to place ovulation, and does ovulation come before or after that peak?

Part 5. Clinical correlation

1 min

TSH 8.4 with a normal free T4

1 min

A patient has a TSH of 8.4 mIU per L, above the usual 0.4 to 4.0, and a free T4 of 1.1 ng per dL, which is inside the reference range. TSH is released by the anterior pituitary, and thyroid hormone feeds back on the pituitary and the hypothalamus. A small change in T4 makes a much larger change in TSH.

Say which end of this axis is failing, and say how the two numbers together tell you that.

Explain why the pituitary hormone leaves its reference range before the thyroid hormone does.

Use your standard curve to say what a sensitive assay is, and why that matters for measuring TSH.

Look back at your thinking

1 min

Before the cortisol and ACTH activity, which patient did you expect to be hardest to place, and why?

What did the data show, and what did you have to change about the axis you had drawn in your head?

Name the one thing about feedback that is now clearer than it was, and say what made it clear.

Turn it in

1 min

Save this page as a PDF with the button below. Photograph your standard curve and both pages of the hormone cycle sheet. If you worked on the printed worksheet, scan or photograph every page instead. Combine everything into one PDF and upload it to the Week 8 lab assignment in Canvas. Due Sunday, November 1 at 10:00 pm.

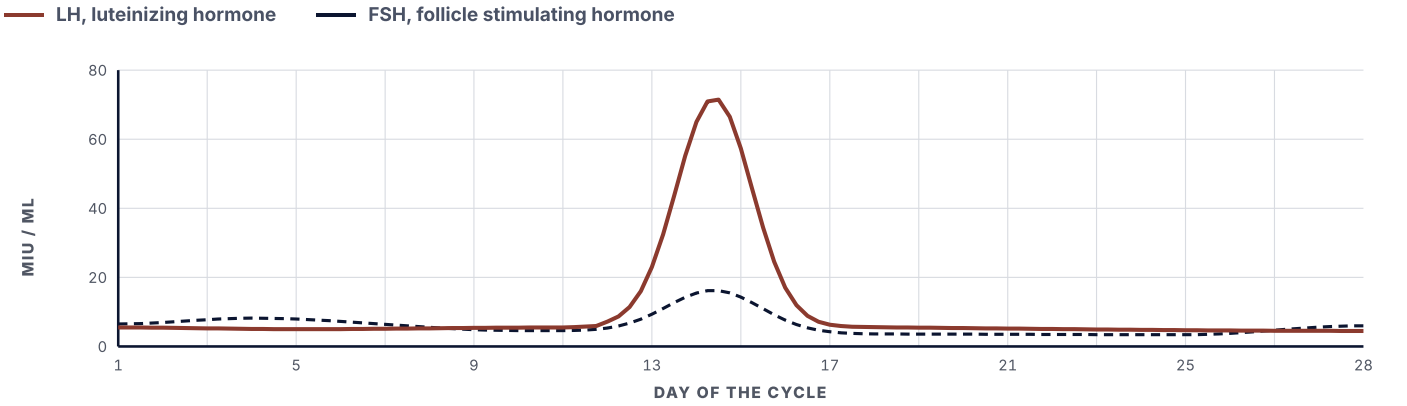
Week 8 lab, Part 4: reading the hormone cycle

FALL 2026
DR. SHARILYN RENNIE

NAME _____ DATE _____

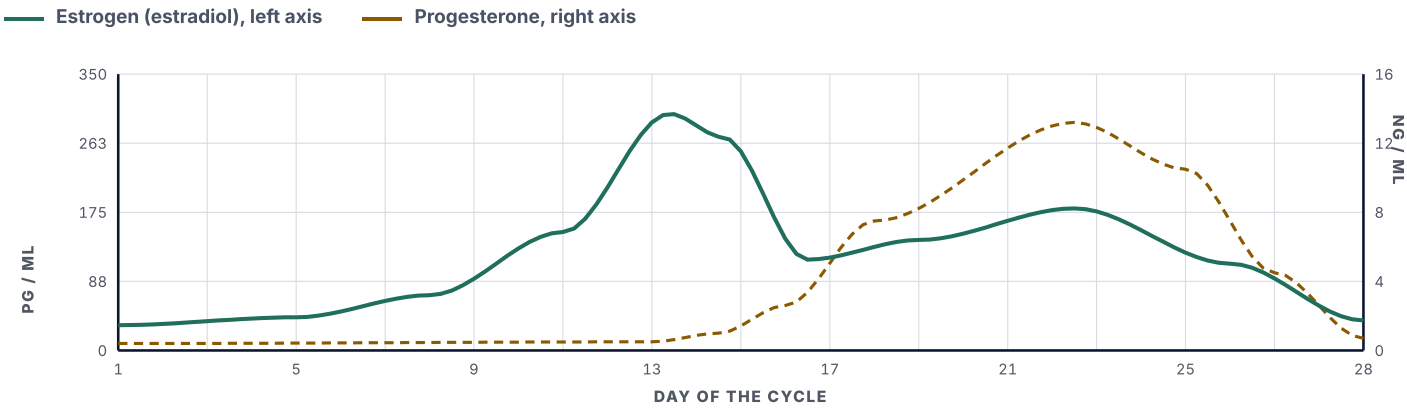
1 The pituitary hormones, one 28 day cycle

Day 1 is the first day of bleeding. Both hormones are plotted on the same axis, in mIU per mL.



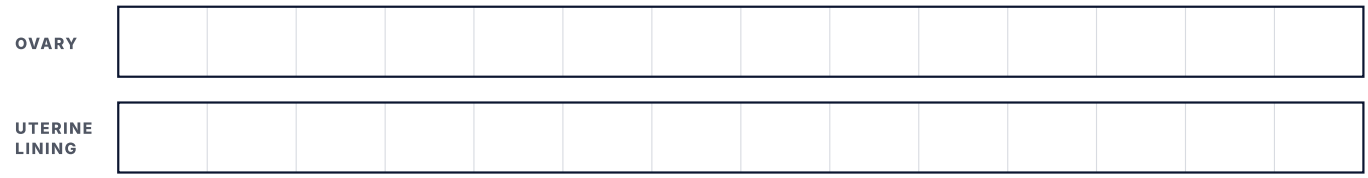
2 The ovarian hormones, same 28 days

Two different scales. Estrogen reads off the left axis in pg per mL. Progesterone reads off the right axis in ng per mL.



3 Label the phases yourself

Same 28 days, lined up under the graphs. Draw the boundaries, write the phase names, and put the day numbers on the boundaries.



DUE SUNDAY 10:00 PM

Week 8 lab, Part 4, page 2: what the graph is telling you

FALL 2026
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4 **Mark the day of ovulation on both graphs, then say how you decided.**

Draw one vertical line straight down through both. Do not guess from memory. Point at the thing on the graph that told you.

5 **Which peaks first, estrogen or LH? By about how many days?**

Read both peaks off the day axis and write the two numbers down before you answer. The order matters more than you think.

6 **Use that order. Which one is causing the other, and how do you know from this graph alone?**

You already know estrogen usually turns the pituitary down. Here it does the opposite. Say what changed.

7 **Name what is happening in the ovary at each of these four points.**

One line each. Say which structure is doing it.

Estrogen climbing, days 7 to 12	
The LH spike, day 13	
Progesterone climbing, days 15 to 21	
Everything falling, days 25 to 28	

8 **Follow the fall at the end of the cycle, step by step.**

Start with the structure that stops working around day 24. End with what the person notices on day 28.

[ ] → [ ] → [ ] → [ ]

9 **Now change one thing. An egg is fertilized on day 15.**

Go back to graph 2 and redraw the progesterone curve from day 22 onward with a dashed line. Then explain below what is holding it up, and what would happen to the uterine lining if it dropped anyway.

10 **What do you know now that you did not know before you started this sheet?**

Three or four sentences in your own words. Say what the graphs showed you that a list of hormones could not.